

Project Summary

Overview. Reinforcement learning (RL) offers a powerful paradigm for sequential decision making under uncertainty, with celebrated successes in complex settings such as Go and Atari games. However, its deployment in operations research (OR) applications remains limited despite the central role of sequential decision making in settings such as supply chain systems, energy grid management, and inventory control. Existing RL methods are often poorly aligned with these systems because they treat the environment as an arbitrary black-box process. This project develops RL methods for *Markov decision processes with exogenous inputs* (Exo-MDPs), a model class that isolates uncertainty in an exogenous stochastic processes, such as demand, arrivals, renewable energy generation, or supplier disruptions, while exploiting the known dynamics that determine how decisions and exogenous inputs affect the system state. The PI proposes three research directions aimed at developing the theoretical and algorithmic foundations of RL for Exo-MDPs:

1. *Learning with observed exogenous inputs:* The PI will characterize the sample complexity, regret, and computational complexity of learning in Exo-MDPs when the exogenous inputs are observed. Whereas black-box RL algorithms scale with the complexity of the full system, this direction will exploit the Exo-MDP representation to design algorithms whose guarantees scale with the complexity of the exogenous stochastic process.
2. *Learning under partial observability:* The PI will develop methods for Exo-MDPs when the exogenous inputs cannot be observed directly and must be inferred from censored or incomplete feedback, such as sales data that only partially reveal underlying demand. This direction will identify how partial observability changes the statistical difficulty of learning and will design algorithms that balance decision quality with information acquisition.
3. *Structure-aware and interpretable policy design:* The PI will develop theoretically grounded gradient-based tools for optimizing policies in Exo-MDPs. This direction will also use large language models to propose interpretable, OR-inspired policy classes that can be optimized and evaluated within the Exo-MDP framework.

These research directions are tightly integrated with the PI's educational plan through the development of ORSuite, an open-source platform for simulator design and benchmarking of reinforcement learning algorithms in operations. This tool will include operational simulators and benchmark algorithms for settings such as inventory control, scheduling, and resource allocation. The package will be integrated into undergraduate and graduate courses at Northwestern.

Intellectual Merit. This project develops a new structure-aware foundation for reinforcement learning in operational systems. By introducing Exo-MDPs as a unifying framework, the project will characterize when and how known dynamics and exogenous uncertainty can be exploited to improve statistical efficiency, computational tractability, and policy design. The resulting theory and algorithms will advance reinforcement learning for operations while contributing broadly to sequential decision making, online learning, and data-driven optimization.

Broader Impacts. The project will broaden access to research and education at the interface of operations research and machine learning through course modules, open educational materials, and student-led development of ORSuite, an open-source platform for operational simulators and reinforcement learning benchmarks. These activities will be integrated into IEMS 408: *Decision Making in Dynamic Learning Environments*, IEMS 351: *Reinforcement Learning*. The project will also develop IEMS 496: *Academic Street Smarts*, and will support outreach through the IDEAL *Get Ready for Research* program and public educational content.